

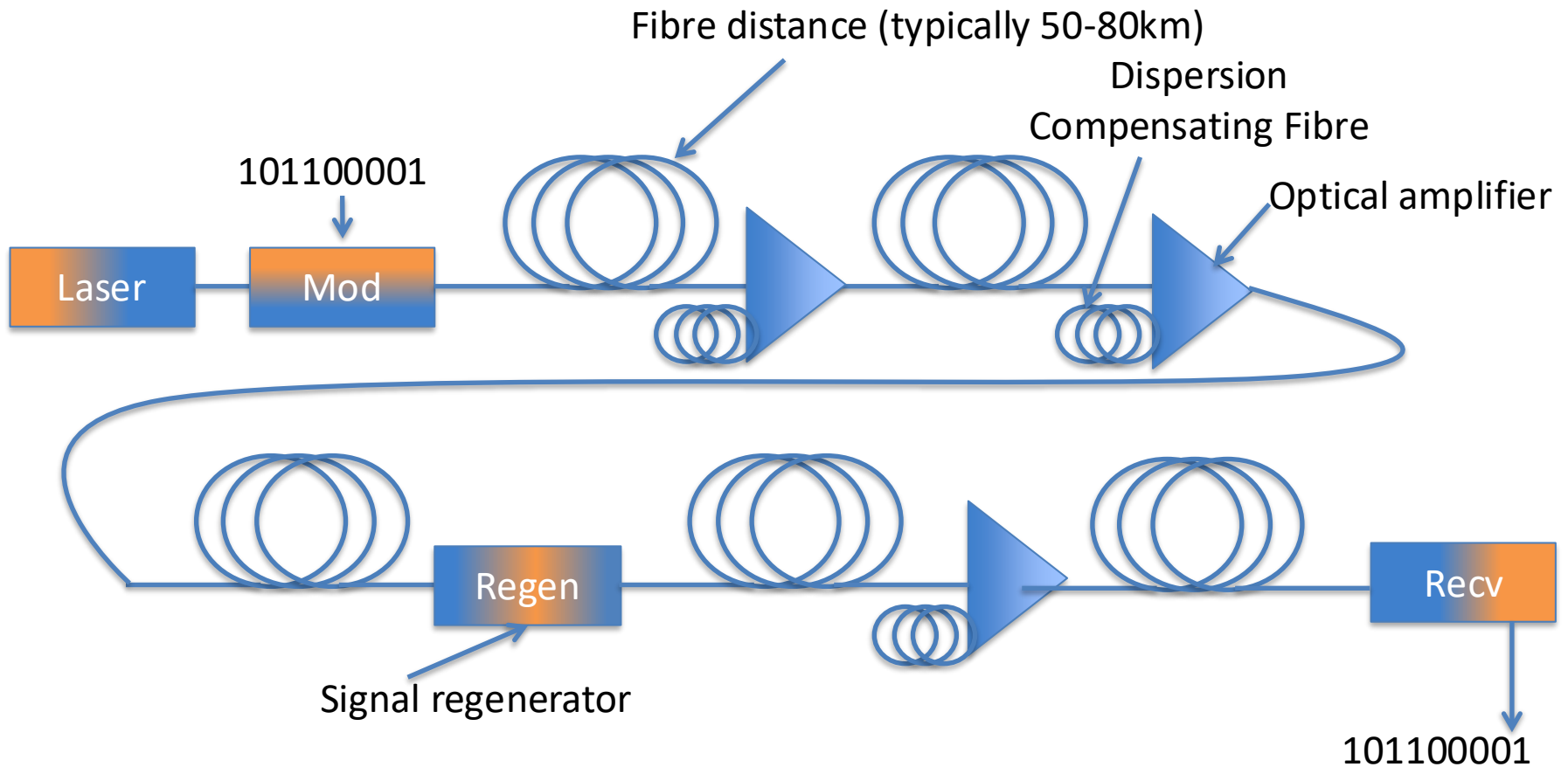
Optical link design

Optical link design

- When designing an optical link I usually have a BER requirement that needs to be satisfied
- In order to satisfy the BER, I need to consider the following constraints:
 - 1) Power budget:** $P_{\text{launch}} - \text{Loss (fibre, connectors,...)} - \text{Margin} \geq \text{receiver sensitivity}$
 - power margin=5dB, considers component degradation and other unforeseen events
 - 2) OSNR:** if power budget is satisfied I also need to satisfy OSNR, so that I have a Q-factor good enough to achieve target BER
 - 3) Other impairments** such as dispersion (modal, chromatic and polarization) and nonlinearities need to be accounted for or compensated.

For 3) we can use the concept of **power penalty** (P_{penalty}): *the increase in the receiver input power needed to eliminate the degradation in the BER caused by fibre impairments (dispersion and nonlinearities)*

Typical optical link



Power budget calculation for amplified systems

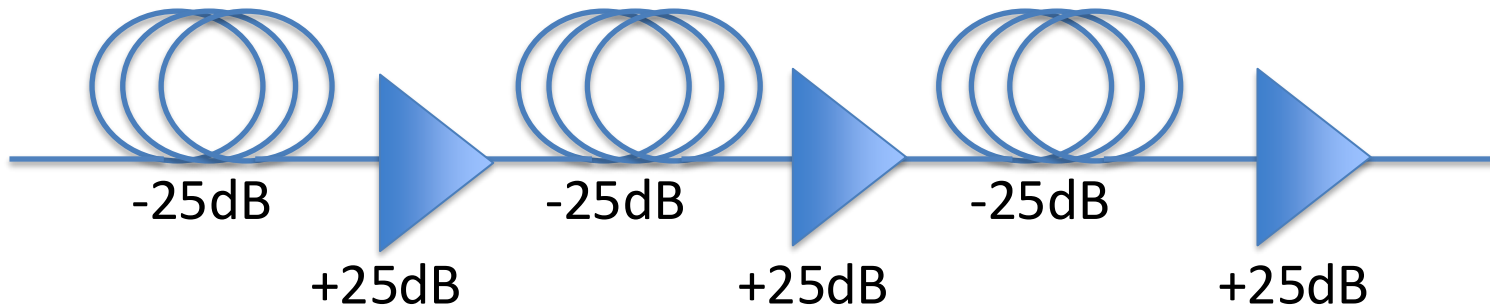
- This is the power budget for an amplified system
- The objective is still to make sure that the received power is $\geq$ **receiver sensitivity**
- I sum up amplification gains and subtract losses on a link span and the system margin (M)

$$P_{\text{recv}} = P_{\text{launch}} - L_1 + G_1 - L_2 + G_2 \dots - L_n + G_n - M \geq \text{receiver_sensitivity}$$

- L_i are the losses of each span between two amplifiers. They will include fibre loss, and loss due to additional equipment (e.g., multiplexers, connectors, dispersion compensating fibre, optical switches...)
- Usually spans are all the same length, the amplifier gains G_i tend to be all equal and they should compensate for the loss L_i preceding the amplifier (again, this will include fibre loss, connectors loss, DCF fibre loss, and other components that might be in the span)

Chain of amplifiers

- For long distances, multiple optical amplifiers are used in cascade
- Each amplifier needs to provide enough gain to overcome the loss of its previous span



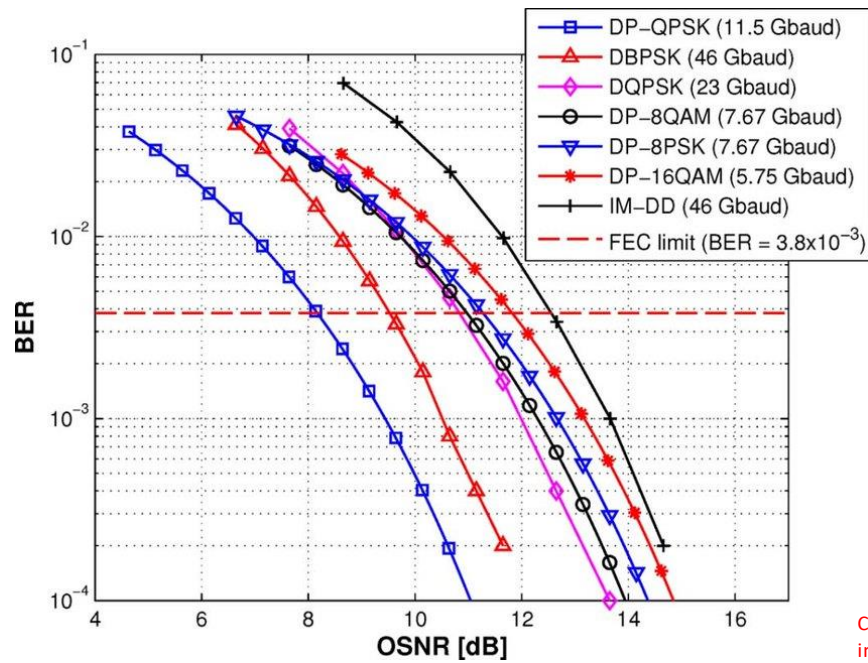
- High gain in the amplifiers increase the ASE noise. Distance between amplifiers can be reduced to decrease the amount of ASE in the system
- Since each amplifier adds noise, the OSNR deteriorates.
- In long-haul systems, before the OSNR becomes too low, the signal needs to be regenerated.

Noise in optical systems

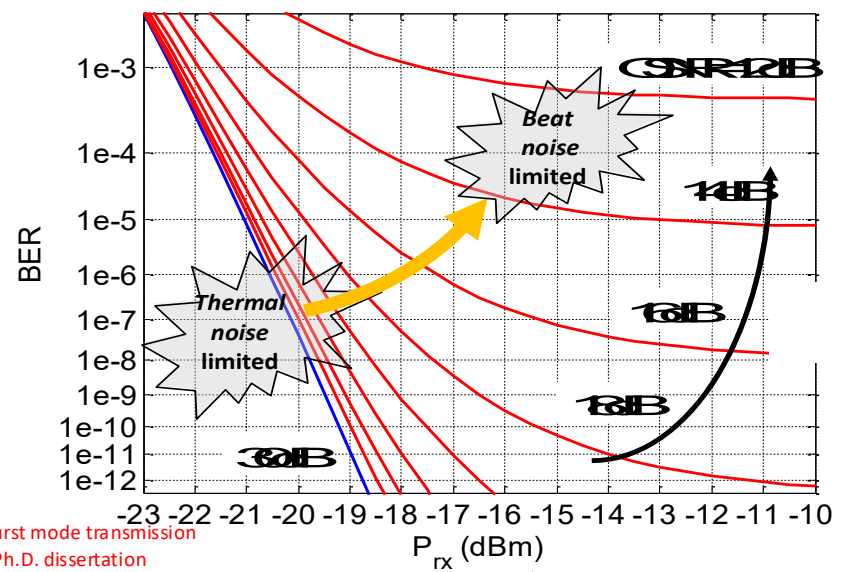
- In optical systems there are three main types of noise
- **Amplified Spontaneous Emission (ASE):** due to optical amplifiers, which besides amplifying the signal carrying information also introduces additional noise.
- **Thermal noise:** due to the random thermal-driven motion of electrons at the receiver, which sums up to the signal carrying information
- **Shot noise:** due to the quantistic nature of an optical signal, the photons carrying the information arrive at the detector with random arrival times

BER and OSNR

- The optical signal-to-noise ratio, is the relation between the power of the signal and the power of the noise at the receiver.
- The optical noise is due to the presence of optical amplifiers
- The higher the OSNR value, the easier will be to determine whether a received bit has a 0 or 1 value.
- The curves depend on the modulation format.. The closer are the symbols the higher the probability of error

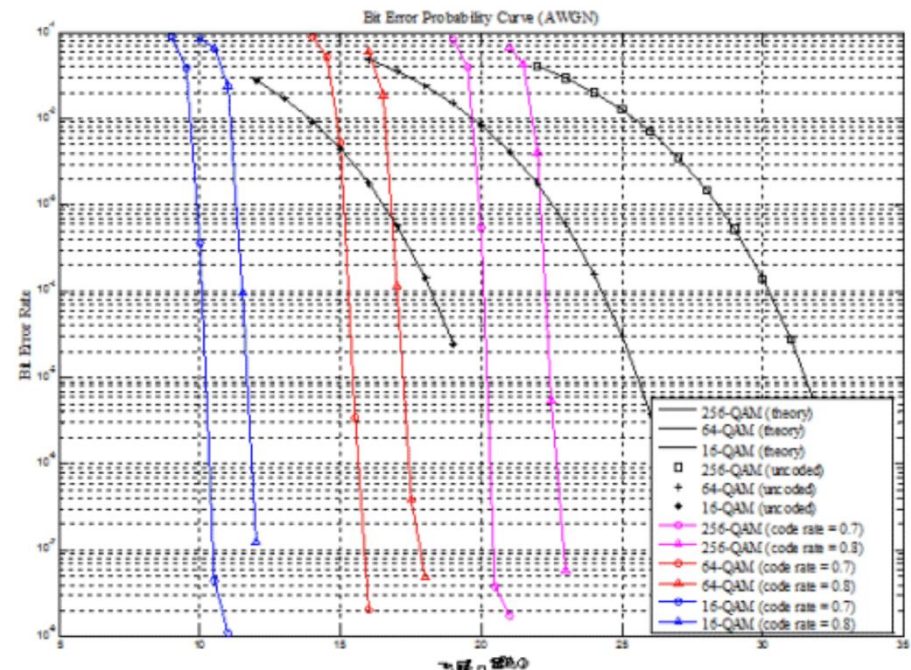
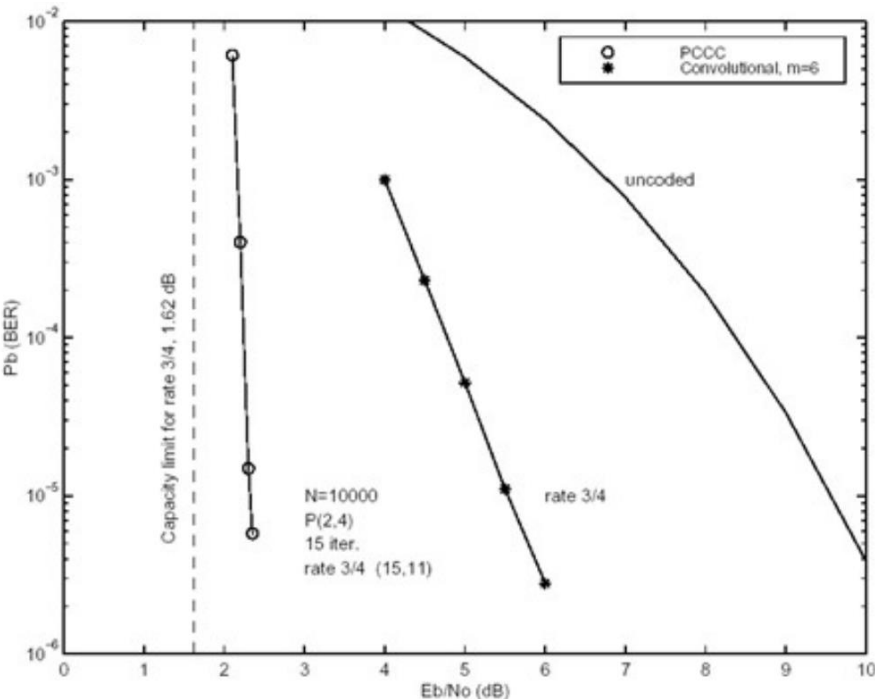


If there are no amplifiers then the optical noise will be low and the limiting factor will be the electrical thermal noise at the receiver



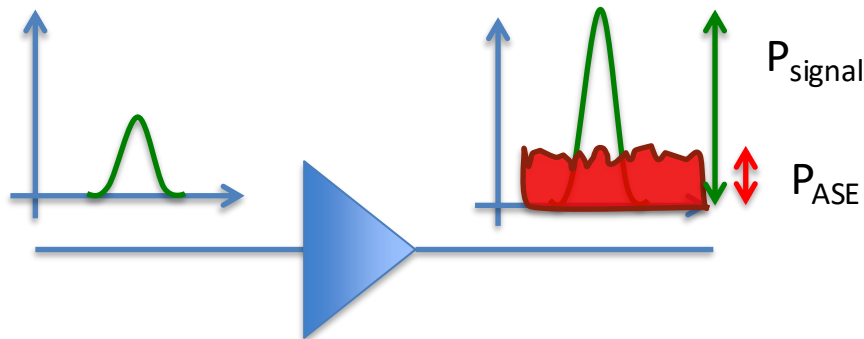
Error correction

- Error correction codes are widely used in telecommunications.
- Typically turbo codes are used in optical communications, which are very close to the Shannon limit and have a very steep curve.
- The maximum BER that a code can correct is called the Forward Error Correction threshold.



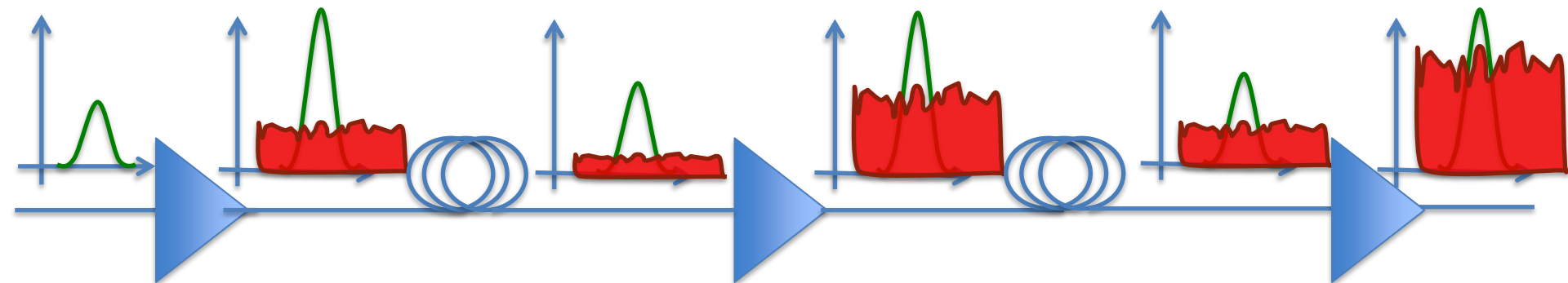
Optical amplifiers and noise

- Optical amplifiers can amplify all incoming optical signals in one fibre.
- However they generate noise, by adding due to This generates a noise called Amplified Spontaneous Emission (ASE)
- Each time the signal is amplified more noise is added
- This phenomenon can be captured by the Optical Signal to Noise Ratio. $OSNR = P_{\text{signal}} / P_{\text{ASE}}$ (calculated over a reference bandwidth)



OSNR for a chain of amplifiers

- The noise figure (NF) is a measure of the noise introduced in the system by an amplifier
- It's measured in dB and depends on the amplifier type, its gain,...
- An average working value for EDFAs can be $NF=5-6\text{dB}$
- **Typically amplifiers are spaced 80Km apart, but this can change**
- **Reducing their spacing increases the system OSNR**
- In a chain of amplifiers, each additional amplifier amplifies both incoming signal and incoming noise, plus it will add its own noise
- After each amplifier the OSNR decreases and **this is permanent, it cannot be recovered!**



Design OSNR calculation

- For a **chain** of n amplifiers the OSNR is:

$$\text{OSNR}_{\text{design}} = P_{\text{TOP[dBm]}} - \text{NF}_{\text{ampl[dB]}} - G - 10\log_{10}(N_{\text{Ch}}) - 10\log_{10}(n) + 58_{\text{[dB]}}$$

- P_{TOP} is the EDFA output power in dBm, NF the noise figure of the EDFA
- G is the amplifier gain, which is typically set to the span loss = $\alpha_{\text{[dB/km]}} L_{\text{span[km]}}$
- N_{Ch} is the number of channels (i.e., the EDFA output power is shared across N channels)
- You can see from this that reducing the amplifiers spacing increases the OSNR
- n is the number of amplifiers in the link
- 58 is a constant related to the central wavelength of the spectrum
- Notice that the OSNR increases with shorter span lengths

What are margins?



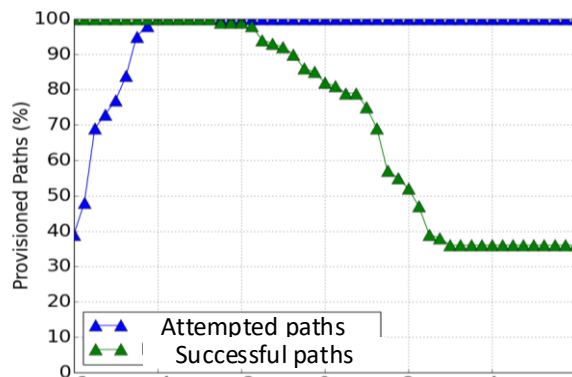
You are close to the cliff edge:

- The closer you get the better (the more efficient you are)
- How safe is it to get closer to the edge, when you can't see it clearly?

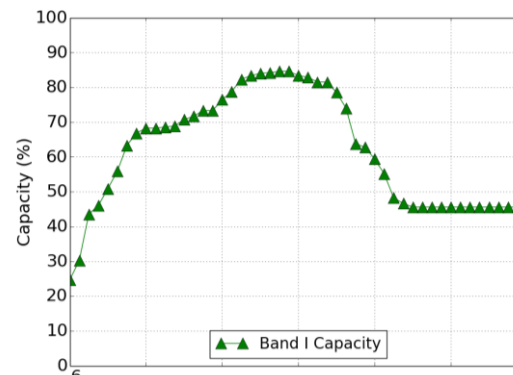
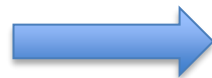
Margin = safety distance (signal to noise ratio) you decide to keep, not being sure where you stand with respect to the edge

In optical communications the margin is the distance to the FEC (Forward Error Correction) Threshold

Effect of use of different margins



Network capacity



10 dB difference makes the difference between a 100G and 400G channel

Application of Margins to OSNR

$$\text{Average OSNR}_{\text{design}} = P_{\text{TOP[dBm]}} - NF_{\text{ampl[dB]}} - G - 10\log_{10}(N_{\text{Ch}}) - 10\log_{10}(n) + 58_{\text{[dB]}}$$



$$\text{Average OSNR}_{\text{Nominal}} = \text{OSNR}_{\text{Design}} - \text{ROADM}_{\text{Penalties}}$$



$$\text{Average OSNR}_{\text{Commissioning}} = \text{OSNR}_{\text{Nominal}} - \text{Supplier_Margins}$$



Crosstalk and filtering penalties



Variations in fiber loss, amplifier power, etc.



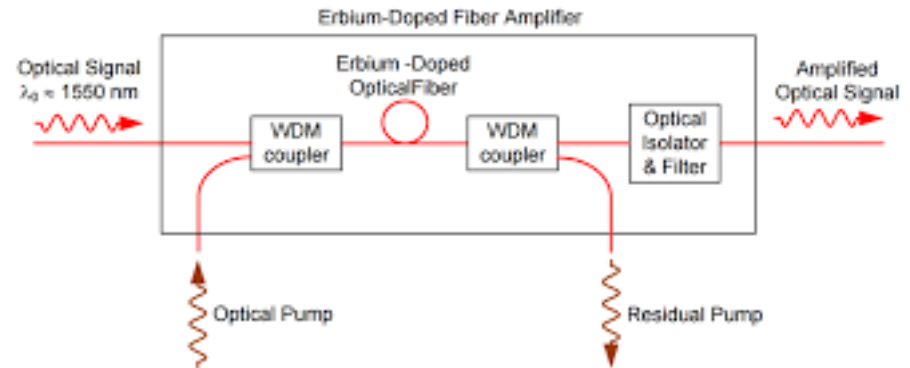
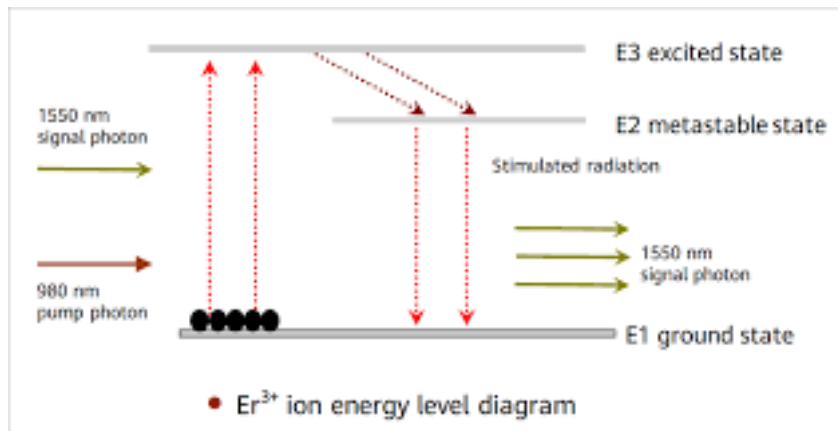
Worst OSNR_{Commissioning} ➔ The worst OSNR across all channels

The Generalised OSNR also includes impairments (noise) due to

nonlinear interference (NLI)
$$\frac{1}{GOSNR} = \frac{1}{OSNR_{ASE}} + \frac{1}{OSNR_{NLI}}$$

Power dynamics in transmission networks

- This is how an Erbium Doped Optical Amplifier works
- A “pump” at 980 nm is used to excite electrons to a higher energy state
- The Erbium doping creates metastable states in the fibre
- As signal photon pass through the fibre, they are amplified by the stimulated emission of radiation effect (like in LASERs)

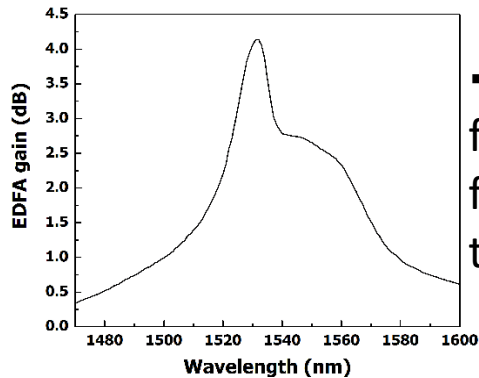


Amplifier operating modes

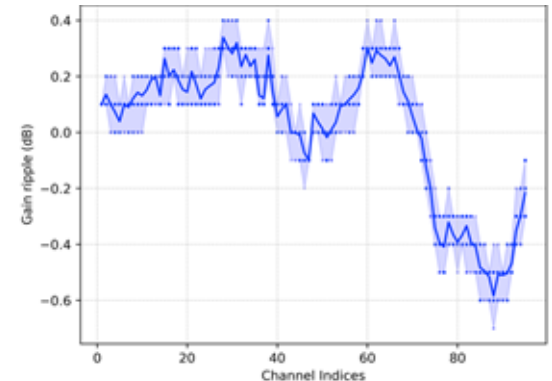
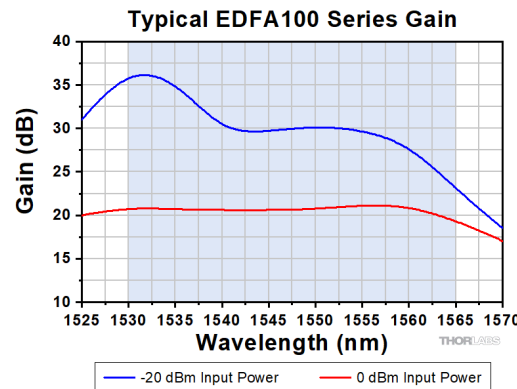
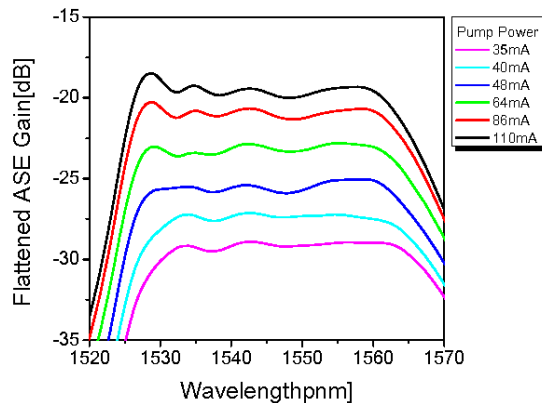
- When I add more channels into the amplifiers, the same energy is shared among more channels.
- Amplifiers can be set to work in Automatic Power Control (APC) or Automatic Gain Control (AGC) mode.
 - APC: the systems controls the pump power to keep the amplifier output constant. If I add more channels, in order to keep the output the same, each channel receives less gain (and vice-versa)
 - AGC (used in all WDM systems): the systems controls the pump power to keep the amplifier gain constant across all channels
 - ➔ A control loop measure the difference between overall output power and overall input power (i.e., the gain), and keeps that constant
 - ➔ If I add channels then the output power is higher, so that each channel has the same gain (and vice-versa).
 - ➔ It makes sense in a WDM system, as the aim is for each channel to gain the same amount of energy that was lost during propagation in the previous span.

Amplifier gain

- The problem of power dynamics is caused by the fact that the gain is not flat across the spectrum, but changes depending on: which other channels are on, pump power, input power. Temperature, etc.



➔ This is the gain before flattening: today flattening filters are applied directly in the amplifier



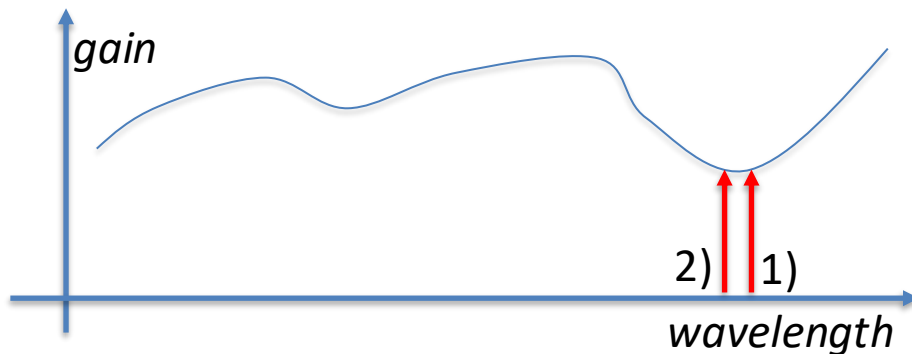
Data collected from different EDFAs (same model)

➔ Even after flattening, the gain varies depending on operating conditions

The issue with non-flat gain

- Non-flat gain causes the problem that channels are amplified differently

==> Notice that the amplifier only has one parameter to play with: the pump power

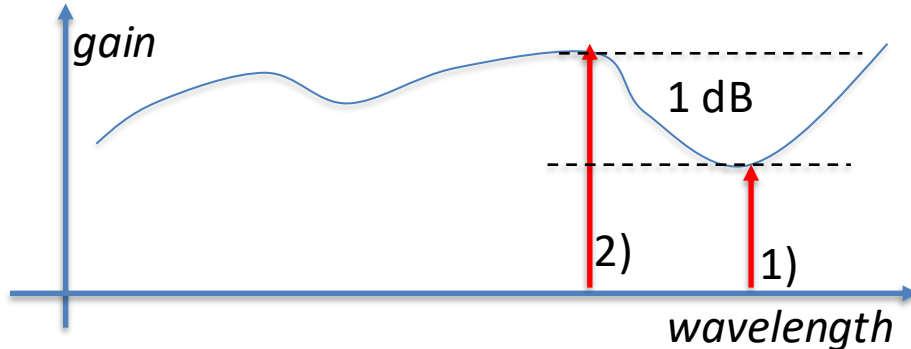


- Consider that channel 1) is in the system and I add channel 2)
- The gain is set to 16 dB and the channels both have -15 dBm power when entering the amplifier

- Channel 1) comes in with -15 dBm and gets out with 1 dBm (gain of 16 dB)

First case: If Channel 2) is in a place of the spectrum with similar gain to Channel 1), then AGC will add pump power say of 10mA so that the overall gain is the same. Since Channel 1) and 2) receive similar gain, their output power will be of 1 dBm for both.

The issue with non-flat gain



Second case: If Channel 2) is in a place of the spectrum with different gain Channel 1), e.g., +1 dB in the figure, then AGC will add pump power less power, say of 5mA, because Channel 2 sees a higher gain (otherwise, with 10mA pump increase the output gain would be higher)

➔ The result is that Channel 2) will see an output power of 1.5 dBm, while Channels 1 will see 0.5 dBm (the overall gain is still 16 dB, but is not shared equally among the channels).

This effect continues as the signals propagate through multiple amplifiers.

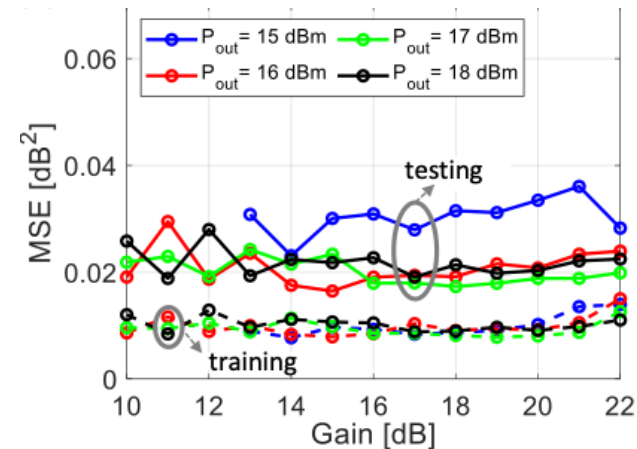
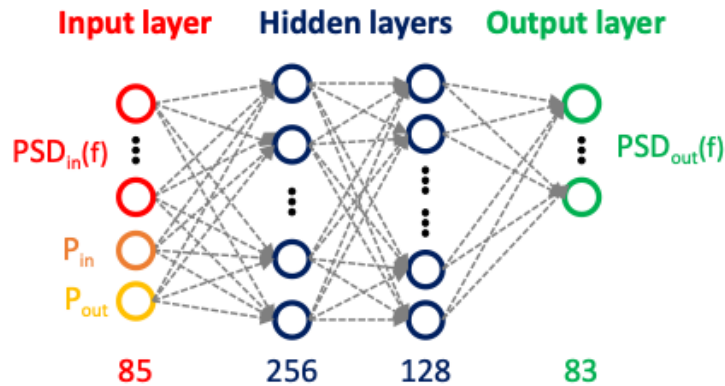
- So, if I add one channel in the system, some of the existing channel might see a drop in power and in OSNR.
- This makes it difficult

Solutions

- Channels are equalized every few spans, so that the difference in power they accumulate is reset (however any loss in OSNR remains)
- In addition, operators might bring the signal up slowly, i.e., add the new channel with a very low power and see how the system reacts, then gradually increase channel power, while monitoring all other channels
- Also some guidelines:
 - If a signal is added near an existing one, there is lower probability of power excursion
 - If a system is already highly loaded (i.e., many existing channels), then adding one more signal has less effect
 - Remember that the amplifier averages the gain across all channels, so the disturbance caused by one more signal is spread across all other signals, reducing the effect on each individual signal

Novel AI-based solution

- The main issue with the gain responses is that they are not known and are difficult to derive mathematically as there is strong dependency on the fabrication process
- However, by collecting many data points over many configuration (dependency on which channels are on gives combinatorial complexity), it is possible to use machine learning to build an accurate model of the amplifier.



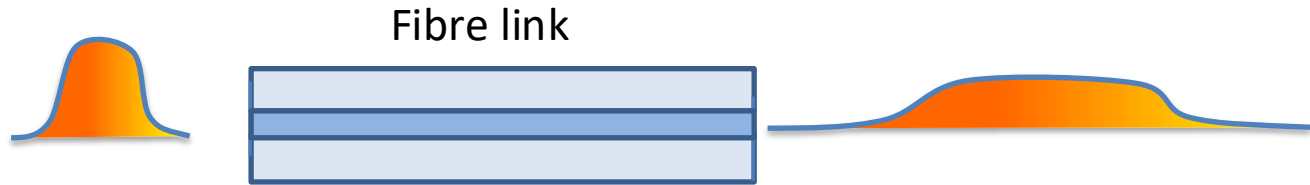
Signal Regeneration

- Before the OSNR drops below the target BER we need to regenerate the signal:
 - you terminate the signal
 - calculate the BER (and make sure it's lower than your target BER)
 - Re-modulate the signal for your next fibre span (like if it was a new signal)
 - For the new span you calculate a new OSNR as you did for the previous span
- **If you regenerate the signal at multiple points the total BER will be the sum of the BER of the individual spans. Remember to take this into account!**
- Regeneration is needed if the OSNR of the total link is too low. If the OSNR is good, but the link budget is low then you might only amplify the signal

Important: *in a WDM system if I decided to put regenerators, I need to put one for every wavelength* → **implies high cost!**

- **For this reason regenerators are only used when absolutely necessary. Often operators sacrifice the signal capacity (i.e., baud rate and modulation rate) to achieve longer distance without regeneration.**

Dispersion



- The spread of the optical pulse in time ΔT is given by:

$$\Delta T = D \cdot L \cdot \Delta \lambda$$

- D is the dispersion coefficient [ps/Km/nm]
- L is the fibre length [Km]
- $\Delta \lambda$ is the signal bandwidth [nm]

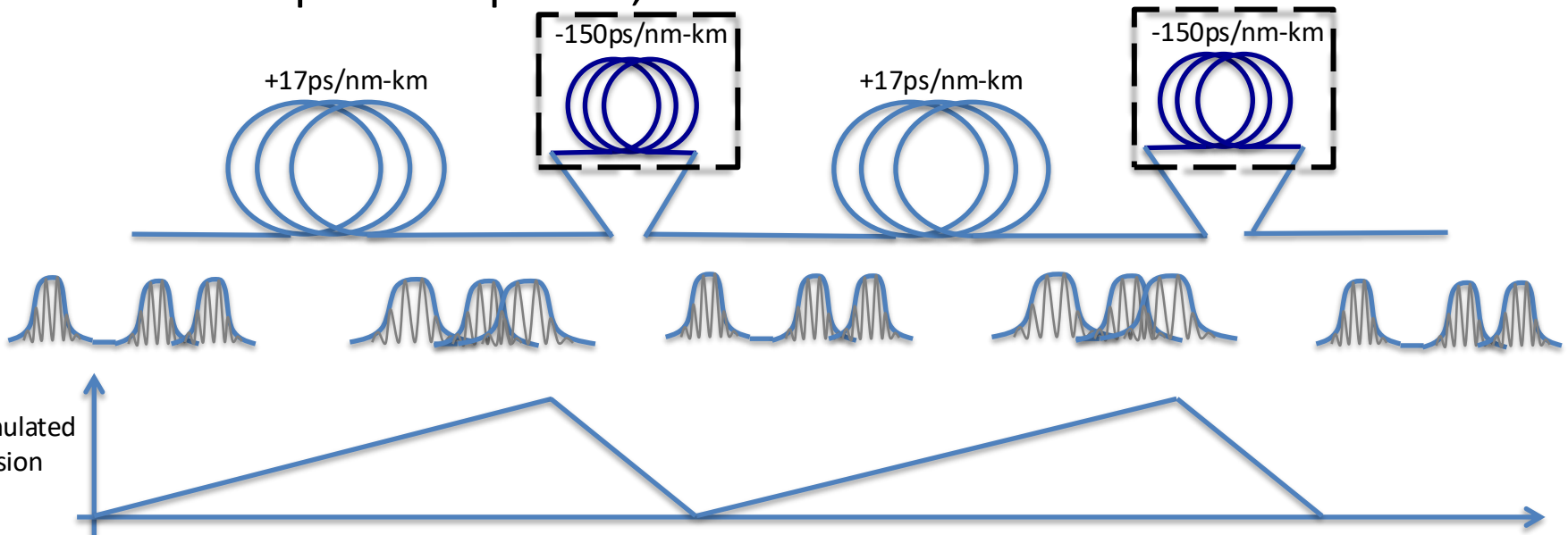
- For each modulation and data rate there's a maximum tolerable dispersion, for which Inter Symbol Interference (ISI) can be tolerated:

$$D \cdot L = \frac{\Delta T}{\Delta \lambda} [ps / nm] < Limit$$

- I have 2 possibilities:
 - If the dispersion is smaller than the dispersion limit, don't do anything and accept the impairment due to dispersion
 - Otherwise, compensate the dispersion to eliminate its effects

Dispersion compensation (used for non-coherent systems)

- If the dispersion accumulated is larger than the limit, it needs to be compensated
- A typical way of compensating chromatic dispersion consists on adding a length of fiber with opposite dispersion characteristic, called Dispersion compensating Fibre (DCF)
- If done with DCF, compensation can be done at the end of the link. However it is usually distributed over multiple stages and co-located with the optical amplifiers, to reduce nonlinear effects

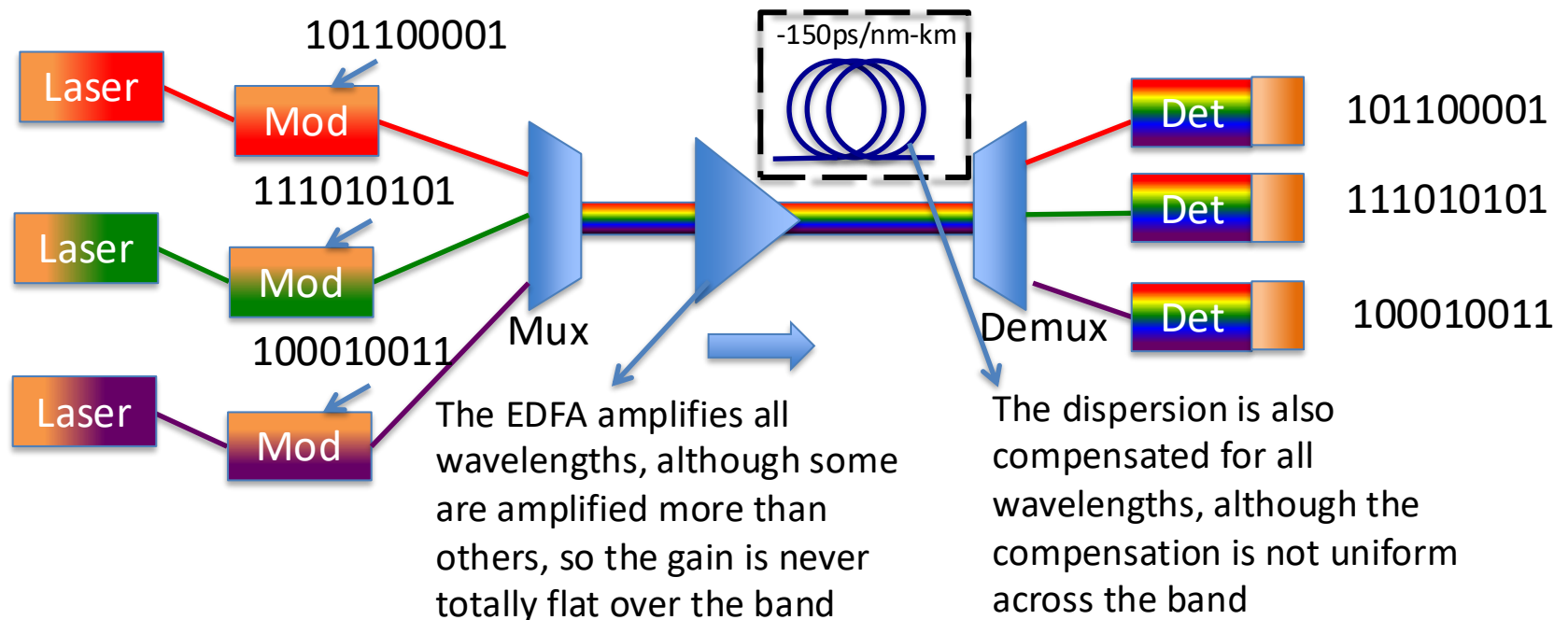


Dispersion compensation

- Dispersion compensating fibre can compensate dispersion on multiple wavelengths at the same time, although the compensation is not perfect along the whole spectrum
- **Notice that it will introduce a loss, which is usually larger than in standard fibre. This needs to be calculated in the power budget and OSNR calculations!**
- With coherent systems, electronic equalization can be used at the receiver to compensate for dispersion, however
 - The higher the data rate and transmission distance the more processing power and memory is required
 - Coherent receivers are more expensive than direct-detection ones

Wavelength Division Multiplexing (WDM)

- Like in (electric) wired and wireless communication different frequency bands can be used to create different communication channels
- Different wavelengths are individually modulated and sent over the fiber

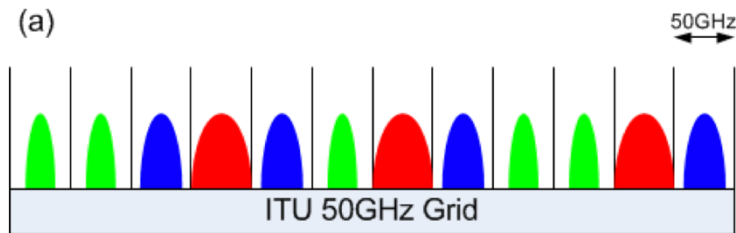


WDM

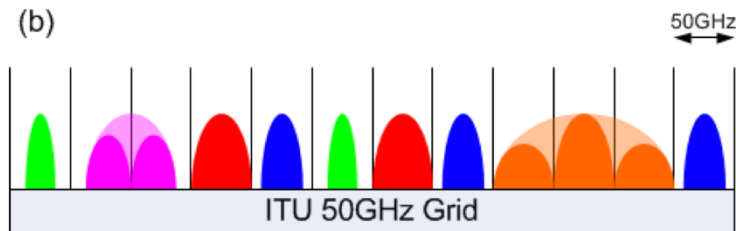
- The ITU-T (International Telecommunications Union – Telecommunication Standardization Sector) has defined a grid for wavelength spacing in the fiber.
- Multiple grids are defined:
- Coarse WDM (CWDM - ITU-T G.694.2):
 - 20 nm spacing: 18 wavelengths in band from 1271 to 1611 nm
 - in C band (1530-1565nm) → 2 wavelengths (1531, 1551)
 - In L band (1565-1625nm) → 3 wavelengths (1571, 1591, 1611)
- Dense WDM (DWDM - ITU-T G.694.1)
 - 100 GHz (~0.8nm): C band - 40 channels
 - 50 GHz (~0.4nm): C band - 80 channels, L band 80 channels
 - 25 GHz, 12.5 GHz not used yet in practical developments

Flex-Grid

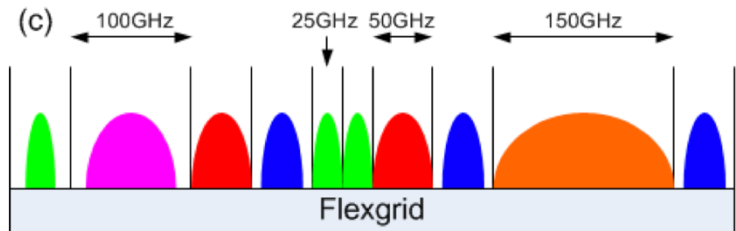
- Recently flexgrid systems have been developed. The optical switches can select dynamically slots of different bandwidth (typically a multiple of 6.25 GHz)



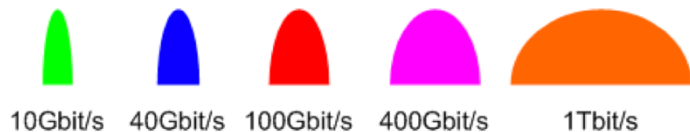
Signals need to stay within the 50GHz filter, but this limits the speed to 100Gb/s



If higher rate signals are used, they will occupy higher bandwidth and will be "notched" by the WDM filters



FlexGrid allows to adapt the filter size, so that higher baud rate signals can be accommodated in the system



A big advantage of a coherent system is that:

- A. It's cheaper to manufacture
- B. Does not need a modulator to transmit information
- C. Dispersion can be dynamically adjusted for link length
- D. The signal can be modulated at higher baud rate